Chapter 2

Introduction to probability

2.1 Introduction

Probability theory is the science that studies random phenomena, that is, experiments whose outcome depends on chance. We cannot predict the result of a die roll; but we know that if we roll the die a large number of times, a 6 will come up about once in six. A probability **measures** how likely an event is, by a number between 0 and 1.

Historically, three approaches to probability followed one another:

- the classical approach: when all outcomes are equally likely, the probability is the ratio of the number of favourable outcomes to the number of possible outcomes (Section 2.5.4);
- the frequentist approach: the probability of an event is the limit of its relative frequency when the experiment is repeated a very large number of times (Section 2.5.5);
- the axiomatic approach (Kolmogorov, 1933): a probability is a function that satisfies a few simple axioms (Section 2.5). This is the framework adopted in this course; the two previous approaches become special cases of it.

2.2 Random experiments and events

Definition 2.1 (Random experiment)

A random experiment is an experiment whose possible outcomes are all known, but whose result cannot be predicted with certainty: repeated under the same conditions, it may give different results.

Speaking of a random experiment means choosing to focus only on the possible outcomes and forgetting the conditions of the experiment. In what follows, ξ denotes a random experiment.

Definition 2.2 (Outcome)

A possible result of the experiment ξ is called an outcome (or elementary result) and is denoted ω.

Definition 2.3 (Sample space)

The set of all possible outcomes is called the sample space (or universe) and is denoted Ω:

$$\Omega = \{\omega_1, \omega_2, \ldots, \omega_n\}. \tag{2.1}$$

Example 2.1 *(Die and coin)*

1. We roll a die and observe the result: $\Omega = \{1, 2, 3, 4, 5, 6\}$, $\text{Card}(\Omega) = 6$.
2. We toss a coin twice (H = heads, T = tails): $\Omega = \{HH, HT, TH, TT\}$, $\text{Card}(\Omega) = 2 \times 2 = 4$. The order matters: HT (heads then tails) and TH are two different outcomes.

Example 2.2 *(DNA sequence)*
We read two successive nucleotides of a DNA sequence at random. Each nucleotide is one of the four bases A, C, G, T, and the order matters: by Chapter 1, there are $4^2 = 16$ outcomes.

$$\Omega = \{AA, AC, AG, AT, CA, CC, CG, CT, GA, GC, GG, GT, TA, TC, TG, TT\}.$$

Example 2.3 *(The same draw, three sample spaces)*
In a factory, three machines M_1, M_2, M_3 produce 300 rods of the same size every day. One makes brass rods, the other two make steel rods. At the end of the day, the rods are mixed and one is drawn at random.

a) If we are interested in the rod itself (its actual size), there are 300 outcomes: $\text{Card}(\Omega_a) = 300$.

b) If we are interested in the machine that made it: $\Omega_b = \{M_1, M_2, M_3\}$.

c) If we are interested in the material: $\Omega_c = \{\text{brass}, \text{steel}\}$.

The same draw leads to three different sample spaces: the sample space depends on **what we observe**.

Remark 2.1 (The three types of sample space)

1. Finite sample space: a finite number of outcomes (the die, the coin, the DNA).
2. Countably infinite sample space: the outcomes can be numbered like the integers. For example, we toss a coin until heads comes up and count the number of tosses: $\Omega = \{1, 2, 3, \ldots\} = \mathbb{N}^*$.
3. Continuous sample space: the outcomes fill an interval of $\mathbb{R}$. For example, we measure the time a computer runs without failure: $\Omega = [0, +\infty[$.

In the first two cases the sample space is discrete, and we will define discrete probability distributions on it; in the third, we will define continuous distributions.

Definition 2.4 (Event)

An event is a subset of Ω, denoted $A, B, C, \ldots$: it is a set of outcomes. We say that the event A occurs if the observed outcome ω belongs to A.

Example 2.4

1. We roll a die. The events $A =$ "get an even number" and $B =$ "get at least 4" are $A = \{2, 4, 6\}$ and $B = \{4, 5, 6\}$. If the die shows 4, both A and B occur; if it shows 5, only B occurs.
2. For the running time of a computer, the event "the computer runs for at least a time t_0" is the interval $A = [t_0, +\infty[$.

2.2.1 Special events

- An elementary event $\{\omega\}$ contains a single outcome. With a die, the elementary events are $\{1\}, \{2\}, \ldots, \{6\}$.
- The impossible event, denoted $\emptyset$, contains no outcome: it never occurs. With a die, "get a 7" is impossible.
- The certain event, denoted Ω, contains all outcomes: it always occurs. With a die, "get a number between 1 and 6" is certain.

2.3 Operations on events

Since events are sets, we combine them with set operations. Each one has a translation in everyday language.

Definition 2.5 (Union)

The union of A and B, denoted $A \cup B$, is the event "A or B": it occurs if **at least one** of the two occurs.

$$\omega \in A \cup B \iff \omega \in A \text{ or } \omega \in B. \tag{2.2}$$

Definition 2.6 (Intersection)

The intersection of A and B, denoted $A \cap B$, is the event "A and B": it occurs if **both** occur.

$$\omega \in A \cap B \iff \omega \in A \text{ and } \omega \in B. \tag{2.3}$$

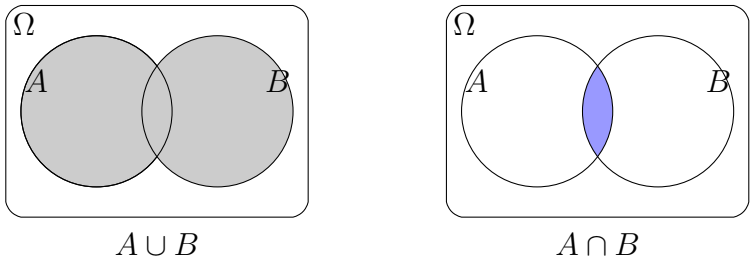


Figure 2.1: Union (left) and intersection (right) of two events

Example 2.5
With the die, $A = \{2, 4, 6\}$ (even) and $B = \{4, 5, 6\}$ (at least 4):

$$A \cup B = \{2, 4, 5, 6\} \text{ (even } \textbf{or} \text{ at least 4)}, \qquad A \cap B = \{4, 6\} \text{ (even } \textbf{and} \text{ at least 4)}.$$

Definition 2.7 (Mutually exclusive events)

Two events A and B are mutually exclusive (or disjoint) if they cannot occur at the same time: $A \cap B = \emptyset$.

Definition 2.8 (Complementary event)

The complement of A, denoted $\bar{A}$ or A^c, is the event that occurs if and only if A does not occur:

$$\bar{A} = \{\omega \in \Omega : \omega \notin A\}. \tag{2.4}$$

We always have $A \cup \bar{A} = \Omega$ and $A \cap \bar{A} = \emptyset$.

Definition 2.9 (Inclusion)

The event A implies the event B if the occurrence of A forces the occurrence of B. We say that A is included in B and write $A \subset B$.

Definition 2.10 (Difference)

The difference of A and B, denoted $A\backslash B$, is the event "A and not B":

$$\omega \in A\backslash B \iff \omega \in A \text{ and } \omega \notin B, \qquad \text{so} \quad A\backslash B = A \cap \bar{B}. \tag{2.5}$$

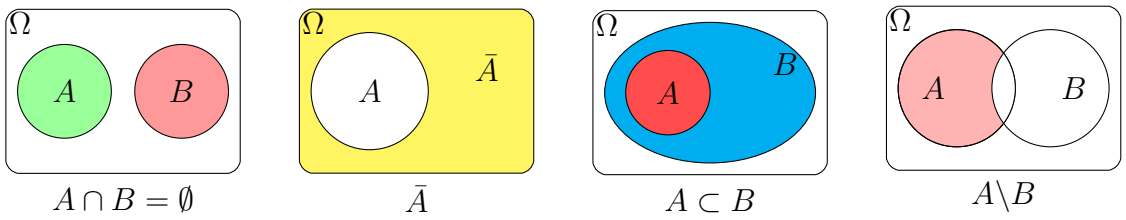


Figure 2.2: Mutually exclusive events, complement, inclusion and difference

Example 2.6
With the die and the same events $A = \{2, 4, 6\}$ and $B = \{4, 5, 6\}$:

- $\bar{A} = \{1, 3, 5\}$: "get an odd number";
- $A\backslash B = \{2\}$: "even but less than 4";
- $\{6\} \subset A$: getting 6 implies getting an even number;
- $\{1\}$ and A are mutually exclusive.

Remark 2.2 (Mutually exclusive is not complementary)

Two complementary events are mutually exclusive, but the converse is false: "get 1" and "get 2" are mutually exclusive without being complementary. For $B = \bar{A}$, we need both $A \cap B = \emptyset$ **and** $A \cup B = \Omega$.

Proposition 2.1 (De Morgan's laws)

For all events A and B:

$$\overline{A \cup B} = \bar{A} \cap \bar{B} \qquad \text{and} \qquad \overline{A \cap B} = \bar{A} \cup \bar{B}. \tag{2.6}$$

Proof. $\omega \in \overline{A \cup B}$ means that ω is neither in A nor in B, that is $\omega \in \bar{A}$ and $\omega \in \bar{B}$. The second identity follows by applying the first to $\bar{A}$ and $\bar{B}$, then taking complements. □

Example 2.7 *(The two disks)*
A server has two disks. Let $A =$ "disk 1 fails" and $B =$ "disk 2 fails". Then:

- "at least one disk fails" is written $A \cup B$;
- "no disk fails" is written $\bar{A} \cap \bar{B} = \overline{A \cup B}$ (De Morgan);
- "only disk 1 fails" is written $A \cap \bar{B} = A \backslash B$.

Table 2.1 summarises the correspondence between the language of probability and that of sets.

Language of probability	**Set notation**
A or B (at least one of the two)	$A \cup B$
A and B (both)	$A \cap B$
not A (complement of A)	$\bar{A}$
A and B are mutually exclusive	$A \cap B = \emptyset$
A implies B	$A \subset B$
A and not B	$A \backslash B = A \cap \bar{B}$
impossible / certain event	$\emptyset$ / Ω

Table 2.1: Vocabulary of events

2.4 Partition of the sample space

Definition 2.11 (Complete system of events)

A family $(A_1, A_2, \ldots, A_n)$ of events forms a complete system of events (or a partition of Ω) if:

1. $A_i \neq \emptyset$, for every $i = 1, \ldots, n$;
2. $A_i \cap A_j = \emptyset$, for all $i \neq j$ (the A_i are pairwise mutually exclusive);
3. $\bigcup_{i=1}^{n} A_i = \Omega$.

In other words, during the experiment, **exactly one** of the events A_i occurs.

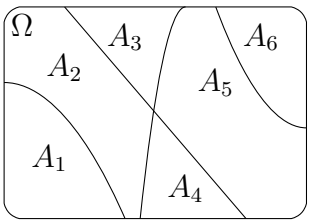


Figure 2.3: A complete system of six events

Example 2.8
We roll a die.

1. $A_1 = \{1, 2\}$, $A_2 = \{3, 4\}$, $A_3 = \{5, 6\}$ form a complete system: they are non-empty, disjoint, and their union is Ω.
2. $B_1 = \{1, 2, 3\}$, $B_2 = \{3, 4, 5, 6\}$ do not form a complete system, because $B_1 \cap B_2 = \{3\} \neq \emptyset$.
3. For every event A other than $\emptyset$ and Ω, the family $(A, \bar{A})$ is a complete system. This case will come up often.

2.5 Probability

2.5.1 Measurable space

Definition 2.12 (Power set)

The power set of Ω, denoted $\mathcal{P}(\Omega)$, is the set of all subsets of Ω, hence of all events. It always contains $\emptyset$ and Ω. If $\text{Card}(\Omega) = n$, then $\text{Card}\left(\mathcal{P}(\Omega)\right) = 2^n$ (Chapter 1).

Example 2.9
For $\Omega = \{1, 2, 3\}$, we get $2^3 = 8$ events:

$$\mathcal{P}(\Omega) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \Omega\}.$$

When Ω is infinite (in particular continuous), a probability cannot always be assigned to every subset of Ω. We then restrict ourselves to a "reasonable" family of events, called a σ-algebra.

Definition 2.13 (σ-algebra)

A σ-algebra $\mathcal{F}$ on Ω is a subset of $\mathcal{P}(\Omega)$ such that:

1. $\Omega \in \mathcal{F}$;
2. if $A \in \mathcal{F}$, then $\bar{A} \in \mathcal{F}$ (closed under complement)
3. if $A_1, A_2, \ldots \in \mathcal{F}$, then $\bigcup_{n\geq 1} A_n \in \mathcal{F}$ (closed under countable union)

The pair $(\Omega, \mathcal{F})$ is called a measurable space.

Example 2.10
For $\Omega = \{1, 2, 3\}$, the family $\mathcal{F} = \{\emptyset, \{1\}, \{2, 3\}, \Omega\}$ is a σ-algebra: it contains Ω, the complement of each of its elements ($\overline{\{1\}} = \{2, 3\}$, $\bar{\Omega} = \emptyset$, ...), and the union of any two of its elements. The power set $\mathcal{P}(\Omega)$ is also always a σ-algebra: the largest one. When Ω is finite, we usually take $\mathcal{F} = \mathcal{P}(\Omega)$.

2.5.2 Axiomatic definition

Definition 2.14 (Probability)

Let $(\Omega, \mathcal{F})$ be a measurable space. A probability (or probability measure) is a function $P : \mathcal{F} \longrightarrow [0, 1]$ such that:

1. $P(\Omega) = 1$;
2. for every sequence $(A_i)_{i\geq 1}$ of **pairwise mutually exclusive** events,

$$P\left(\bigcup_{i\geq 1} A_i\right) = \sum_{i\geq 1} P\left(A_i\right). \tag{2.7}$$

In particular, if $A \cap B = \emptyset$, then $P(A \cup B) = P(A) + P(B)$. The triple $(\Omega, \mathcal{F}, P)$ is called a probability space.

Example 2.11 *(The biased coin)*
We toss a biased coin: heads comes up twice as often as tails. Let $P(\{T\}) = p$, so $P(\{H\}) = 2p$. The events $\{H\}$ and $\{T\}$ are mutually exclusive and their union is Ω, so

$$P(\{H\}) + P(\{T\}) = P(\Omega) = 1 \implies 3p = 1 \implies P(\{T\}) = \frac{1}{3}, \quad P(\{H\}) = \frac{2}{3}.$$

2.5.3 Properties

Theorem 2.1 (Properties of a probability)

Let A and B be two events of a probability space $(\Omega, \mathcal{F}, P)$. Then:

1. $P(\emptyset) = 0$;
2. $P(\bar{A}) = 1 - P(A)$;
3. if $A \subset B$, then $P(A) \leq P(B)$;
4. $P(A \backslash B) = P(A \cap \bar{B}) = P(A) - P(A \cap B)$;
5. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$;
6. if $(A_1, \ldots, A_n)$ is a complete system of events, then $P(B) = \sum_{i=1}^{n} P(B \cap A_i)$.

Proof. Each property is obtained by writing an event as a union of mutually exclusive events, then applying (2.7).

1. $\Omega = \Omega \cup \emptyset$ with $\Omega \cap \emptyset = \emptyset$, so $P(\Omega) = P(\Omega) + P(\emptyset)$, hence $P(\emptyset) = 0$.
2. $A \cup \bar{A} = \Omega$ and $A \cap \bar{A} = \emptyset$, so $P(A) + P(\bar{A}) = P(\Omega) = 1$.
3. If $A \subset B$, then B is the union of the two mutually exclusive events A and $B \backslash A$. So $P(B) = P(A) + P(B \backslash A)$, which is at least $P(A)$ since $P(B \backslash A) \geq 0$.
4. $A = (A \cap B) \cup (A \backslash B)$, a disjoint union, hence $P(A) = P(A \cap B) + P(A \backslash B)$.
5. $A \cup B = (A \backslash B) \cup (A \cap B) \cup (B \backslash A)$, three pairwise disjoint events. With property 4:
$$P(A \cup B) = \big[P(A) - P(A \cap B)\big] + P(A \cap B) + \big[P(B) - P(A \cap B)\big] = P(A) + P(B) - P(A \cap B).$$
6. $B = B \cap \Omega = (B \cap A_1) \cup \cdots \cup (B \cap A_n)$, and the $B \cap A_i$ are pairwise disjoint because the A_i are. □

Remark 2.3 (Why subtract $P(A \cap B)$?)

In $P(A) + P(B)$, the outcomes of $A \cap B$ are counted **twice**: once in A and once in B. We therefore remove them once. If A and B are mutually exclusive, we recover $P(A \cup B) = P(A) + P(B)$. For three events, the same reasoning gives the inclusion–exclusion formula:

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C).$$

Example 2.12 *(Python and C)*
In a class, 40 % of the students know Python (A), 30 % know the C language (B) and 15 % know both. A student is chosen at random.

- Probability that the student knows at least one of the two languages: $P(A \cup B) = 0.40 + 0.30 - 0.15 = 0.55$.
- Probability that the student knows neither: $P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B}) = 1 - 0.55 = 0.45$.
- Probability that the student knows Python but not C: $P(A \setminus B) = 0.40 - 0.15 = 0.25$.

2.5.4 Equally likely outcomes

Let ω be an outcome. The probability $P(\{\omega\})$ of the elementary event $\{\omega\}$ is called an elementary probability. When Ω is finite, a probability is entirely determined by its elementary probabilities: $P(A) = \sum_{\omega \in A} P(\{\omega\})$.

Proposition 2.2 (Uniform probability)

Let Ω be a finite sample space. If all outcomes have the same probability (we say they are equally likely), then

$$P(\{\omega\}) = \frac{1}{\text{Card}(\Omega)} \quad \text{for every } \omega \in \Omega,$$

and, for every event A:

$$P(A) = \frac{\text{Card}(A)}{\text{Card}(\Omega)} = \frac{\text{number of favourable outcomes}}{\text{number of possible outcomes}}. \tag{2.8}$$

Proof. Let $N = \text{Card}(\Omega)$ and p the common probability of the outcomes. The N elementary events are mutually exclusive and their union is Ω, so $Np = 1$, that is $p = \frac{1}{N}$. An event A is the union of its $\text{Card}(A)$ elementary events, hence $P(A) = \text{Card}(A) \times \frac{1}{N}$. □

Remark 2.4 (When can formula (2.8) be used?)

This formula is valid only if the outcomes are **equally likely**. Phrases such as "at random", "fair die", "fair coin" or "balls identical to the touch" allow us to assume it. Computing a probability then amounts to **counting**, with the tools of Chapter 1.
Warning: with two identical dice, the 21 unordered results are **not** equally likely: a double such as $(3, 3)$ corresponds to a single pair, whereas $\{1, 2\}$ corresponds to two pairs, $(1, 2)$ and $(2, 1)$. We therefore always work with the 36 ordered pairs, which are equally likely.

Example 2.13 *(Two dice)*
We roll two fair dice. What is the probability that the sum of the faces equals 7? The sample space consists of the 36 equally likely pairs (i, j). The favourable outcomes are $(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)$, hence

$$P(\text{sum} = 7) = \frac{6}{36} = \frac{1}{6}.$$

Example 2.14 *(Three dice)*
We roll three fair dice: $\text{Card}(\Omega) = 6^3 = 216$.

1. *Exactly one* 6. We choose the die that shows 6 (3 ways); the other two show one of the 5 other numbers (5^2 ways):
$$P = \frac{3 \times 5^2}{216} = \frac{75}{216} = \frac{25}{72} \simeq 0.35.$$
2. *At least one* 6. We use the complement, "no 6", which has $5^3 = 125$ outcomes:
$$P = 1 - \frac{125}{216} = \frac{91}{216} \simeq 0.42.$$

Rule of thumb: for "at least one", use the complement.

Example 2.15 *(Brown eyes)*
A class has 10 girls and 20 boys; half of the boys and half of the girls have brown eyes. A person is chosen at random. Let $G =$ "the person is a boy" and $M =$ "the person has brown eyes". There are 15 people with brown eyes, 10 of whom are boys:

$$P(G \cup M) = P(G) + P(M) - P(G \cap M) = \frac{20}{30} + \frac{15}{30} - \frac{10}{30} = \frac{25}{30} = \frac{5}{6}.$$

Example 2.16 *(The urn)*
An urn contains 8 balls identical to the touch: 5 red and 3 white. We draw 3 balls simultaneously. The draw being simultaneous, $\text{Card}(\Omega) = C_8^3 = 56$. To get exactly 2 red balls, we choose 2 red among 5 **and** 1 white among 3:

$$P = \frac{C_5^2 \times C_3^1}{C_8^3} = \frac{10 \times 3}{56} = \frac{15}{28} \simeq 0.54.$$

Example 2.17 *(The lottery)*
In a lottery, 50 tickets are sold, only 2 of which are winning tickets. We buy 4 tickets. The complement of "win at least one prize" is "all 4 tickets lose", that is, they are chosen among the 48 losing tickets:

$$P(\text{win at least one prize}) = 1 - \frac{C_{48}^4}{C_{50}^4} = 1 - \frac{194\,580}{230\,300} \simeq 0.155.$$

2.5.5 Frequency and probability

Definition 2.15 (Relative frequency)

If an event A occurs k times in n repetitions of an experiment, its relative frequency is the ratio

$$f_n(A) = \frac{k}{n}. \tag{2.9}$$

It is sometimes expressed as a percentage.

As the number n of repetitions increases, the frequency $f_n(A)$ settles around the probability $P(A)$: this is the law of large numbers, which will be proved in the chapter on modes of convergence. In practice, the probability of an event is therefore often modelled by its observed frequency on a large sample.

Example 2.18

1. Over 6 000 rolls of a fair die, we expect to see a 6 about 1 000 times: $f_n(\{6\}) \simeq \frac{1}{6}$.
2. If a manufacturer observes that 2 % of the components produced are defective, it will say that a component picked at random is defective with probability 0.02.

2.6 Conditional probability

We roll a fair die: the probability of getting 4 is $\frac{1}{6}$. If we know **in advance** that the result is even, only three outcomes remain, 2, 4 and 6, and the probability of getting 4 becomes $\frac{1}{3}$. Information changes probabilities: this is the subject of this section.

Definition 2.16 (Conditional probability)

Let A and B be two events, with $P(B) \neq 0$. The conditional probability of A given B is the number

$$P(A|B) = P_B(A) = \frac{P(A \cap B)}{P(B)}. \tag{2.10}$$

It is the probability that A occurs when we know that B has occurred: the sample space "shrinks" to B.

Example 2.19

1. We roll a die. Probability of getting an even number given that it is greater than 3: with $A = \{2, 4, 6\}$ and $B = \{4, 5, 6\}$,

$$P(A|B) = \frac{P(\{4, 6\})}{P(\{4, 5, 6\})} = \frac{2/6}{3/6} = \frac{2}{3}.$$

2. An urn contains 10 white marbles, 5 yellow and 10 black. A marble is drawn and we observe that it is not black. The probability that it is yellow is

$$P(Y|\bar{K}) = \frac{P(Y)}{P(\bar{K})} = \frac{5/25}{15/25} = \frac{1}{3},$$

where $K =$ "black", because $Y \cap \bar{K} = Y$: a yellow marble is never black.

Proposition 2.3

Let B be an event with $P(B) > 0$. The function $A \longmapsto P(A|B)$ is a probability on $(\Omega, \mathcal{F})$, called the conditional probability distribution given B.

Remark 2.5

$P(\cdot|B)$ therefore has all the properties of Theorem 2.1. In particular $P(\bar{A}|B) = 1 - P(A|B)$. However, in general, $P(A|\bar{B}) \neq 1 - P(A|B)$: it is the event **before** the bar that may be replaced by its complement, never the one after it.

2.6.1 Multiplication rule

Definition (2.10) can also be read the other way round. It gives the probability that **several** events occur **one after another**.

Proposition 2.4 (Multiplication rule)

If $P(A) \neq 0$ and $P(B) \neq 0$:

$$P(A \cap B) = P(A) \times P(B|A) = P(B) \times P(A|B). \tag{2.11}$$

For three events with $P(A \cap B) \neq 0$:

$$P(A \cap B \cap C) = P(A) \times P(B|A) \times P(C|A \cap B). \tag{2.12}$$

Example 2.20 *(Drawing without replacement)*
An urn contains 5 red balls and 3 white ones. We draw 3 balls one after another without replacement. Let R_i = "the i-th ball is red". After each red ball drawn, one red ball and one ball in total are removed:

$$P(R_1 \cap R_2 \cap R_3) = P(R_1) \times P(R_2|R_1) \times P(R_3|R_1 \cap R_2) = \frac{5}{8} \times \frac{4}{7} \times \frac{3}{6} = \frac{5}{28}.$$

2.6.2 Law of total probability

Theorem 2.2 (Law of total probability)

Let $(A_1, \ldots, A_n)$ be a complete system of events with non-zero probabilities. For every event B:

$$P(B) = \sum_{i=1}^{n} P(B|A_i) \times P(A_i). \tag{2.13}$$

Proof. By property 6 of Theorem 2.1, $P(B) = \sum_{i=1}^{n} P(B \cap A_i)$, and, by the multiplication rule, each term equals $P(B|A_i)\,P(A_i)$. □

Corollary 2.1 (Case of two events)

If $0 < P(B) < 1$, the complete system $(B, \bar{B})$ gives, for every event A:

$$P(A) = P(A|B) \times P(B) + P(A|\bar{B}) \times P(\bar{B}). \tag{2.14}$$

Example 2.21 *(The plants)*
A company builds cars in four plants. The table gives the share of production of each plant U_i and the probability that a car coming out of it is defective (D):

Plant	U_1	U_2	U_3	U_4
$P(U_i)$	0.2	0.3	0.4	0.1
$P(D\|U_i)$	0.05	0.01	0.01	0.02

The plants form a complete system. We read the probability tree (Figure 2.4): we multiply along each branch (multiplication rule) and add the branches that lead to D (law of total probability):

$$P(D) = 0.2 \times 0.05 + 0.3 \times 0.01 + 0.4 \times 0.01 + 0.1 \times 0.02 = 0.010 + 0.003 + 0.004 + 0.002 = 0.019.$$

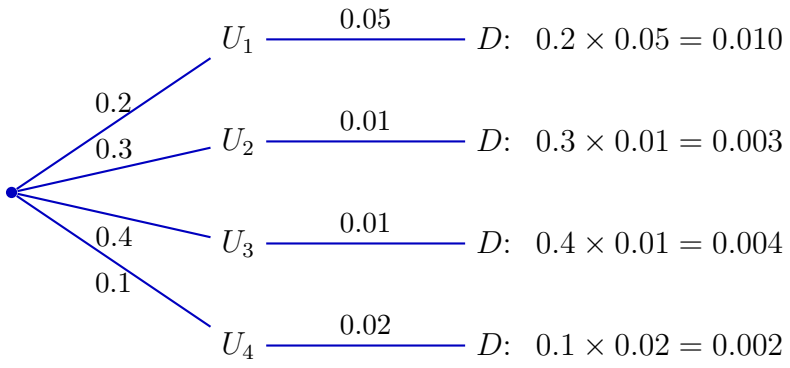


Figure 2.4: Probability tree (only the branches leading to D are shown)

2.6.3 Bayes' formula

We often know the probability of an **effect** given its **cause** ($P(D|U_1)$: the car is defective given that it comes from U_1), and we look for the probability of the cause given the effect ($P(U_1|D)$). Bayes' formula lets us go back from the effect to the cause.

Theorem 2.3 (Bayes' formula)

Let $(A_1, \ldots, A_n)$ be a complete system of events with non-zero probabilities, and B an event with $P(B) \neq 0$. For every $j = 1, \ldots, n$:

$$P(A_j|B) = \frac{P(B|A_j) \times P(A_j)}{P(B)} = \frac{P(B|A_j) \times P(A_j)}{\sum_{i=1}^{n} P(B|A_i) \times P(A_i)}. \tag{2.15}$$

Proof. By definition, $P(A_j|B) = \dfrac{P(A_j \cap B)}{P(B)}$. The numerator equals $P(B|A_j)\,P(A_j)$ (multiplication rule) and the denominator is given by the law of total probability (2.13). □

Example 2.22 *(Back to the plants)*
A car is defective. What is the probability that it comes from plant U_1?

$$P(U_1|D) = \frac{P(D|U_1)\,P(U_1)}{P(D)} = \frac{0.05 \times 0.2}{0.019} = \frac{10}{19} \simeq 0.53.$$

Plant U_1 builds only 20 % of the cars, but more than half of the defective cars come from it.

Example 2.23 *(Two machines: a complete example)*
Two machines M_1 and M_2 make mechanical parts. They produce $\frac{1}{3}$ and $\frac{2}{3}$ of the output respectively. Machine M_1 produces 5 % defective parts, machine M_2 produces 6 %. Let $A =$ "the part is made by M_1", $B = \bar{A} =$ "the part is made by M_2" and $D =$ "the part is defective".

1. *A part is drawn from the output. Probability that it was made by M_1:* $p_1 = P(A) = \dfrac{1}{3}$.
2. *A part is drawn from the output of M_1. Probability that it is defective:* $p_2 = P(D|A) = \dfrac{5}{100} = \dfrac{1}{20}$.
3. *Probability that a part comes from M_1 and is defective* (multiplication rule):
$$p_3 = P(A \cap D) = P(D|A) \times P(A) = \frac{1}{20} \times \frac{1}{3} = \frac{1}{60}.$$
4. *Probability that a part is defective.* (A, B) is a complete system; law of total probability:
$$p_4 = P(D) = P(D|A)\,P(A) + P(D|B)\,P(B) = \frac{1}{60} + \frac{6}{100} \times \frac{2}{3} = \frac{5}{300} + \frac{12}{300} = \frac{17}{300} \simeq 0.057.$$
5. *Probability that a defective part was made by M_1* (Bayes):
$$p_5 = P(A|D) = \frac{P(D|A)\,P(A)}{P(D)} = \frac{1/60}{17/300} = \frac{1}{60} \times \frac{300}{17} = \frac{5}{17} \simeq 0.29.$$

Example 2.24 *(Did the student really know?)*
Facing a multiple-choice question with $m = 4$ answers, a student knows the answer with probability $p = 0.6$; otherwise the student guesses and has one chance in 4 of being right. The answer is correct: what is the probability that the student knew it? Let $R =$ "the student knows the answer" and $C =$ "the student answers correctly". With $P(C|R) = 1$ and $P(C|\bar{R}) = \frac{1}{4}$:
$$P(R|C) = \frac{P(C|R)\,P(R)}{P(C|R)\,P(R) + P(C|\bar{R})\,P(\bar{R})} = \frac{1 \times 0.6}{0.6 + \frac{1}{4} \times 0.4} = \frac{0.6}{0.7} = \frac{6}{7} \simeq 0.86.$$
A correct answer is therefore not absolute proof of knowledge: in about 14 % of cases, it is due to chance.

2.7 Independent events

Definition 2.17 (Independence of two events)

Two events A and B are independent if
$$P(A \cap B) = P(A) \times P(B). \tag{2.16}$$
If $P(A) \neq 0$ and $P(B) \neq 0$, this is equivalent to $P(B|A) = P(B)$, or to $P(A|B) = P(A)$: **the occurrence of one does not change the probability of the other**.

Example 2.25
We roll a fair die twice; Ω contains 36 equally likely pairs. The events A = "get 1 on the first roll" and B = "get 6 on the second roll" are independent:

$$P(A \cap B) = P(\{(1,6)\}) = \frac{1}{36} = \frac{1}{6} \times \frac{1}{6} = P(A) \times P(B).$$

Remark 2.6 (Independent is not mutually exclusive)

Do not confuse independent events (the occurrence of one does not change the probability of the other) with mutually exclusive events (the occurrence of one prevents the other). With a single die:

- $A = \{2,4,6\}$ and $\bar{A} = \{1,3,5\}$ are mutually exclusive, but **not** independent: $P(A \cap \bar{A}) = 0 \neq \frac{1}{2} \times \frac{1}{2}$. Knowing that A occurred tells us everything about $\bar{A}$;
- $A = \{2,4,6\}$ and $C = \{1,2\}$ are independent: $P(A \cap C) = P(\{2\}) = \frac{1}{6} = \frac{1}{2} \times \frac{1}{3}$.

Two mutually exclusive events with non-zero probabilities are **never** independent.

Proposition 2.5

If A and B are independent, then A and $\bar{B}$, $\bar{A}$ and B, and $\bar{A}$ and $\bar{B}$ are also independent.

Proof. A is the union of the mutually exclusive events $A \cap B$ and $A \cap \bar{B}$, so

$$P(A \cap \bar{B}) = P(A) - P(A \cap B) = P(A) - P(A)\,P(B) = P(A)\,\big(1 - P(B)\big) = P(A)\,P(\bar{B}).$$

The other two cases follow by swapping the roles of A and B, then applying the result to $\bar{A}$ and B. □

Definition 2.18 (Mutual independence)

The events $A_1, \ldots, A_n$ are mutually independent if, for **every** subfamily $A_{i_1}, \ldots, A_{i_k}$ $(2 \leq k \leq n)$:

$$P(A_{i_1} \cap \cdots \cap A_{i_k}) = P(A_{i_1}) \times \cdots \times P(A_{i_k}). \tag{2.17}$$

For three events A, B, C, they must be pairwise independent **and** satisfy $P(A \cap B \cap C) = P(A)\,P(B)\,P(C)$.

Example 2.26 *(Pairwise is not enough)*
We toss two fair coins. Let A = "tails on the first coin", B = "tails on the second coin" and C = "tails on exactly one of the two coins". Then $P(A) = P(B) = P(C) = \frac{1}{2}$ and

$$P(A \cap B) = P(\{TT\}) = \frac{1}{4}, \qquad P(A \cap C) = P(\{TH\}) = \frac{1}{4}, \qquad P(B \cap C) = P(\{HT\}) = \frac{1}{4}.$$

The three events are therefore pairwise independent. Yet $A \cap B \cap C = \emptyset$ (two tails and exactly one tail at the same time is impossible), so $P(A \cap B \cap C) = 0 \neq \frac{1}{8}$: they are **not** mutually independent.

Proposition 2.6

If A, B and C are mutually independent, then:

- A and $B \cap C$ are independent; A and $B \cup C$ are independent;
- $\bar{A}$, B, C are mutually independent, and so are $\bar{A}$, $\bar{B}$, $\bar{C}$.

Example 2.27 *(Back to the two machines)*
We return to Example 2.23. We take one part made by M_1 and one part made by M_2. Let C = "the part from M_1 is defective" and E = "the part from M_2 is defective". The two machines work separately: C and E are independent.

1. Probability that both parts are defective:

$$p_6 = P(C \cap E) = P(C) \times P(E) = 0.05 \times 0.06 = 0.003.$$

2. Probability that neither is defective. By Proposition 2.5, $\bar{C}$ and $\bar{E}$ are also independent:

$$p_7 = P(\bar{C} \cap \bar{E}) = (1 - 0.05) \times (1 - 0.06) = 0.95 \times 0.94 = 0.893.$$

Example 2.28 *(System in series)*
A computer system consists of 3 components connected in series: it works only if all three work. Each component works with probability 0.9, independently of the others. Let F_i = "component i works". By mutual independence,

$$P(\text{the system works}) = P(F_1 \cap F_2 \cap F_3) = 0.9^3 = 0.729,$$

and the system fails with probability $1 - 0.729 = 0.271$.

Example 2.29 *(The three shooters)*
Three shooters independently aim at the same target; they hit it with probabilities 0.6, 0.3 and 0.8 respectively. What is the probability that the target is hit? Let A_i = "shooter i hits the target". The target is hit if **at least one** shooter hits it. We use the complement, "nobody hits it", and the independence of $\bar{A}_1$, $\bar{A}_2$, $\bar{A}_3$:

$$P(A_1 \cup A_2 \cup A_3) = 1 - P(\bar{A}_1 \cap \bar{A}_2 \cap \bar{A}_3) = 1 - 0.4 \times 0.7 \times 0.2 = 1 - 0.056 = 0.944.$$

The inclusion–exclusion formula gives the same result, but with many more computations.
